Engineering Physics for EEE stream (BPHYE102/202)

QUESTION BANK

MODULE 2 – ELECTRICAL PROPERTIES OF SOLIDS

Conductors

1. Write the assumptions of quantum free electron theory.
2. Define Fermi Energy and Fermi factor. Write expression for both.
3. Discuss the dependence of Fermi factor on temperature and energy levels with graph.
4. Mention the expression for electrical conductivity and explain the terms.

Dielectric properties

5. Distinguish between Polar and non-Polar Dielectrics.
6. What is Dielectric Polarization? Explain different types of polarization mechanisms with suitable diagrams.
7. Write a note on Internal filed in solids.
8. Derive an expression for Clasius–Mossotti equation in dielectric materials.
9. Explain briefly Solid, Liquid and Gaseous dielectrics.
10. Explain the use of dielectrics in transformers.

Superconductivity

11. What are Superconductors? Explain Temperature dependence of resistivity in a superconductor.
12. Write a note on Meissner's effect.
13. Define Critical filed. Explain Temperature dependence of Critical field.
14. Explain BCS theory.
15. Describe Type I and type II superconductors with graph.
16. Write a note on High Temperature superconductivity.
17. What are SQUIDS? Explain the working of DC SQUIDS.
18. Explain the construction and working of MAGLEV vehicle.

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